A Survey on Adversarial Patch Attacks on Target Detection: Transferability and Mitigation Analysis

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***Abstract*—Adversarial patches form a considerable threat for object detection since the localized visual patch can result in either missed detection or localization error whereas the rest of the image is untouched. This paper offers empirical results on the vulnerability of target detection to adversarial patches, their transferability to unseen models, and the possible mitigation of their adverse effects through diffusion methods. Methodology involves the testing of diffusion-generated adversarial candidates using Hyper-YOLO-N as a source model. Next, these patches were transferred to five victim architectures: Faster R-CNN, RT-DETRv2-S, D-FINE-S, YOLOv12-S, and Deformable DETR-R50. 1,500 pairs of patched images with targets as people and cars were used and Attack Success Rate (ASR), confidence reduction and localization variation were measured using the ground-truth based matching criteria with confidence and IoU thresholds of 0.25 and 0.50 respectively. ASR values are 5.61% for Hyper-YOLO-N and 1.17%–3.87% for five target architectures. Moreover, ASR is not sufficient for measuring the effect of attacks as the confidence and IoU of surviving detections varied. Mitigation of the attacks with Automatic DISPATCH pipeline resulted in Detection Recovery Rate (DRR) of 13.33% (recovered two attacked targets) while preserving 13 out of 15 controls (Control Preservation Rate, CPR = 86.67%). An additional experiment included reconstruction of the attacked image region using Stable Diffusion 2.1 Base.**

***Index Terms*—Adversarial patch, object detection, adversarial attacks, transferability, Hyper-YOLO, diffusion models, Stable Diffusion, DISPATCH, adversarial defense.**

I. Introduction

Object detection is one of the fundamental tasks in the field of computer vision, which consists of identifying objects and estimating their location within an image. Modern object detectors are utilized in a variety of fields ranging from autonomous systems, intelligent transportation, surveillance, robotics, and others. The progress in the field of deep learning resulted in the development of efficient detectors like the YOLO detectors, region-based detectors, and transformer-based detectors. However, modern object detectors are vulnerable to specific types of inputs, which are created to mislead a model by manipulating the visual information.

There are several types of adversarial attacks on deep neural networks, which exploit the sensitivity of the models. Among them, adversarial patches are important for object detection because these types of attacks only affect a specific part of the input image and are visible and printable. As opposed to low-level noise spread throughout the image, a patch can be physically printed and attached to any object. Such an attack can mislead an object detector, making it unable to detect an object, classify an existing object incorrectly, or detect something that is not there. Early studies have shown that an adversarial patch can be used for attacking object detectors in realistic situations, such as attacking person detectors in surveillance systems [1]. Later works developed this idea to include adversarial T-shirts, which do not lose their effectiveness when the human body posture changes [2], as well as naturalistic physical patches [3].

The evolution of adversarial patch attacks led to numerous studies related to both digital and physical-world attacks, concerns regarding their transferability, and defense techniques. The current literature focuses predominantly on attacks against various architectures used for object detection, studying different aspects like physical deformations, appearance of patches, generalization, and resistance to perturbations. At the same time, defense-oriented research is oriented towards detecting, segmenting, removing, or neutralizing the patches. In Adversarial YOLO, a patch detection module is embedded in the YOLO detector [4]. Segment and Complete uses patch segmentation and shape completion to remove regions containing adversarial patches detected by this technique [5]. PatchZero conducts pixel-level detection and patch zeroing without requiring retraining of the object detector [6]. The latest approach to addressing adversarial patch impacts in object detection is diffusion-based image regeneration [7].

On the other hand, the object detection architecture itself is constantly evolving. Various changes have been made during several generations of the YOLO architecture and described in recent review works [8], [9]. Hyper-YOLO introduces the use of hypergraph computations to account for higher-order connections between visual features and can be considered an advanced version of the YOLO detector [10]. Studies on adversarial robustness show that the vulnerabilities depend on the architecture and its configuration [11]. Apart from YOLO, there are other architectures for object detection, such as Faster R-CNN, Deformable DETR, RT-DETRv2, D-FINE, and YOLOv12 [12]–[16].

This paper presents a survey of adversarial patch attacks against target detection tasks and its empirical study where

attention is paid to recent developments in adversarial patch attack research and cross-architecture vulnerability. Though there has been a significant amount of progress achieved in terms of research into adversarial patches, the current literature suffers from fragmentation due to the variety of attack types, targeted detector architecture, evaluation metrics, and defenses used. Most previous studies focus on the single type of detector, a certain physical environment, particular adversarial patch attack formulation, and one specific defense technique. Thus, a thorough analysis of the adversarial attacks requires the consideration of how adversarial attacks affect several architectures of object detectors and what metrics should be applied to measure their influence. More specifically, a distinction can be made between complete detection failure and mere confidence or localization changes of the target.

The current paper presents an organization of a survey and empirical analysis concerning the topic of adversarial attacks on target detection using modern object detectors and paying attention to cross-architecture vulnerability. Hyper-YOLO is the source detector architecture in this case, and the existing adversarial images are analyzed with respect to several unseen detector architectures. An evaluation procedure is adopted in order to measure the effect of adversarial patches in a unified way by calculating attack success rate (ASR), confidence shift, and IoU change for every attack-target pair. It allows for detecting complete target-level detection failures and the cases when a target is still detectable while the prediction confidence or localization has changed.

Furthermore, the current paper considers the aspect of mitigation of adversarial patch attacks through the use of diffusion-based restoration. Namely, the current study distinguishes between the problems of automatic localization of the adversarial area and restoration of the target image after such localization.

The main contributions of this paper are as follows:

- An overview of adversarial patches in object detection with regard to digital/physical, rigid/naturalistic forms of patches, transferability, and defense techniques.
- An empirical study of existing Hyper-YOLO adversarial images for multiple architectures of object detectors using a unified framework of evaluation.
- A target-wise investigation using attack success rate, change in confidence, and change in localization based on IoU metric as distinct forms of adversarial effect.
- An analysis of diffusive mitigation method where the task of localization is separated from the restoration task due to known localization diagnostics.

II. Related Work

Studies involving the use of adversarial patch attacks have developed side-by-side with the fast growth of deep object detection frameworks. Previous studies have looked into the vulnerabilities of different types of detectors, construction of digital and physical patches, their robustness against practical modifications, and defenses to detect or remove these adversarial regions. In this part, the relevant literature in adversarial patch attacks for object detection is discussed, while placing this current study in context.

A. Evolution of Object Detection Architectures

As a result of the development of modern detectors, a great variety of architectures have been formed, which use different approaches for feature extraction, localization, and prediction. One of the paradigms that has proved to be very popular in the field of real-time object detection is the YOLO family of algorithms; with each new generation, there are changes to the architecture aimed at increasing its accuracy, computational efficiency, and speed. Current surveys show the evolution from the early versions of YOLO to modern versions and emphasize problems in real-time object detection [8], [9]. At the same time, some research has been conducted to evaluate the adversarial robustness of YOLO-based detectors and to show that the vulnerability properties can be different depending on the architecture of the models [11].

One of the recent versions of the YOLO family, which is called Hyper-YOLO, uses hypergraph computation for modeling higher-order interactions between visual features. This framework offers such architectures as HGC-SCS (Hypergraph Computation Empowered Semantic Collecting and Scattering), MANet (Mixed Aggregation Network), and HyperC2Net (Hypergraph-Based Cross-Level and Cross-Position Representation Network). It should be noted that the architectural characteristics of Hyper-YOLO make it especially relevant to security analysis, including checking whether the effects of adversarial attacks can be observed in detectors with different internal architectures.

Besides approaches based on YOLOs, object detection has developed further via region-based and transformer-based architectures. In Faster R-CNN, the architecture consists of a Region Proposal Network to obtain candidate object regions along with the deep feature extraction process [12]. Deformable DETR employs deformable attention to focus on sparse and informative sampling locations, thus overcoming the problems associated with convergence and spatial processing in conventional DETRs [13]. RT-DETRv2 upgrades real-time transformer detection by making improvements in multi-scale feature sampling and training schemes [14], whereas D-FINE reformulates bounding-box regression as fine-grained distribution refinement to improve localization accuracy [15]. More recent advancements have been made with the release of YOLOv12, which uses an attention mechanism for YOLO-based architectures while still ensuring real-time detection capability [16]. Such diversification in architectures has specific importance in the study of adversarial robustness, as an adversarial impact in one detector is not necessarily the same in other architectures.

B. Adversarial Patch Attacks on Object Detection

Adversarial patches constitute a special type of attack where a small, optimized area within an image is used in order to affect the predictions made by a vision model. In the case of object detection, these kinds of attacks might affect both

the classification and localization processes, thus making the object invisible or changing the detector's response toward the object. Compared to traditional distributed attacks, patch-based attacks are especially relevant in the case of physical-world security since the modified part of the image could be applied in practice to a scene captured by a camera.

Thys et al. investigated patch-based attacks on person detection and showed that such local patterns could help to decrease the possibility of detecting individuals in surveillance scenes [1]. This paper is especially important since the class of person shows a lot of variation in the intra-class data, thus creating an additional difficulty for the attack compared to other objects with a relatively stable appearance. Another advantage of this paper is the consideration of practical problems of real-world cameras.

Afterward, attention was drawn to how physical deformation could affect adversarial patches. Xu et al. created physical adversarial T-shirts that helped to bypass the person detector even in the case of non-rigid deformation caused by a change in human pose [2]. This research also explored an ensemble attack based on the combination of YOLOv2 and Faster R-CNN. This field is especially important because the physical application of the patches means that there are additional transformations caused by the human body apart from those in static images.

Finally, attention was paid to the appearance of such adversarial patches. Hu et al. presented a novel approach for generating physical adversarial patches in terms of their naturalness using a learned image manifold [3]. This work expands the investigation of patch-based attacks in the field of not only attack efficiency but also the visual appearance of the resulting patterns. Overall, all of these works show the evolution of adversarial patches from completely digital patterns to physical attacks considering deformation and naturalness.

C. Cross-Detector Transferability

A central issue in adversarial robustness is how an attack designed for one detector works when applied to another detector. The cross-model transferability issue takes on added importance in practice due to the fact that the attacker does not know anything about the architecture or other internal information of the target detector. Previous studies showed that the effects of attacks can go beyond a particular detector. In the adversarial T-shirt experiment, the optimization was performed for two detectors–YOLOv2 and Faster R-CNN–and thus, constructing an attack for several detector architectures is possible [2]. Also, the investigation of YOLO robustness has shown that attack behavior may be different from one generation and architecture of the model to another [11].

However, the issue of transferability is not limited to the target being completely undetected by another detector. It is possible that the adversarial image retains the target detection but still affects the output score or target localization in some way. Therefore, the evaluation of adversarial patch transferability should take into account different metrics, such as target-level detection validity, attack success rate, score difference, and IoU change.

D. Defenses Against Adversarial Patch Attacks

Along with adversarial patches, another body of defense techniques has been developed in parallel. First of all, several works focus on the explicit detection of the adversarial region and the incorporation of the latter into the detection pipeline. The authors of Adversarial YOLO (Ad-YOLO) added an adversarial-patch class to the YOLO framework that allowed for the detection of both target objects and the adversarial patches themselves [4]. In addition, adversarial training was used to improve recognition of patches of different shapes.

Secondly, another body of work focuses on localizing the patch at the pixel level and performing the patch removal or reconstruction afterwards. In particular, SAC by Liu et al. utilizes the combination of adversarial-patch segmentation and shape completion technique to remove the detected malicious patch [5]. Thus, the main goal of the research was to increase the robustness to unseen shapes of patches, budgets, and even modalities of the attacks. The researchers managed to prove that successful patch localization is a vital element of a universal defense.

Finally, PatchZero tries to implement a different way of restoring the image after being attacked by focusing on the adversarial region localization at the pixel level and substituting the detected region with mean pixel values [6]. Unlike the methods focused on the alteration of the architecture of the detector, PatchZero is concerned with pre-processing the input while keeping the detection pipeline unchanged. All of these works show that good defense against adversarial patches requires two connected elements – patch localization and restoration/neutralization of the patch effect.

E. Diffusion Models for Adversarial Generation and Mitigation

Recently, however, diffusion models have proved to be a powerful generative paradigm for adversarial example generation. Specifically, DiffProtect explores the possibility of generating adversarial examples using diffusion-based generation in the context of facial privacy protection and demonstrates how generative models can generate meaningful adversarial perturbations [17]. Meanwhile, DiffAdvMAP is a diffusion-based paradigm for generating unrestricted adversarial examples and explores the trade-offs between attack strength, naturalness, flexibility, and transferability of the attacks [18]. Recently, there has been work on generating object-level unrestricted adversarial examples through diffusion-model attention to localize semantic object areas while creating adversarial examples [19].

However, despite these efforts that demonstrate the applicability of diffusion models for adversarial example generation, they do not seem to pay much attention to unrestricted adversarial examples or tasks unrelated to adversarial patch defense. One direction related more directly to the research is diffusion-based image restoration for object detection tasks. Ma et al.

introduced DisPatch, an algorithm combining diffusion-based image generation with the localization of adversarial regions based on the difference and selective rectification of those regions in order to address adversarial patches for object detection [7].

F. Research Perspective and Literature Synthesis

First, adversarial attacks specific to certain detectors have been demonstrated for modern detectors, including person detectors. Second, the use of the physical world increases the complexity due to the issues of patch deformation and variability in the environment. The examples include adversarial T-shirts and physical patches. Third, transferability becomes an important security characteristic because adversarial attacks can be generalized across detector architectures. Fourth, a number of possible defenses have been proposed, such as detection of the adversarial image, segmentation, removal, and image restoration.

Despite the large number of papers on the topic, the studies have different source and target detector architectures, ways of generating adversarial images, datasets, criteria for the effectiveness of adversarial attacks, and assumptions about the potential defense. As a result, there is no unified way of evaluating how one set of adversarial images works for multiple detector architectures and at the same time differentiating between the complete failure of a detector and any decrease in the confidence level or localization. The goal of this survey paper is to address these two complementary problems and give an empirical point of view on cross-architecture adversarial transfer.

Unlike other surveys that propose new ways of adversarial patch attack generation for a certain detector architecture, the current paper explores the already generated set of Hyper-YOLO adversarial images and applies it to unseen detectors. In addition, the survey paper evaluates not only the success rate but also paired changes in confidence and Intersection over Union (IoU). Moreover, the analysis of mitigations distinguishes between automatic region localization and patch removal.

III. Problem Formulation and Threat Model

A. Problem Formulation

Denote by x a pristine image containing one or more target objects and by f_s an object detector which in our experiment is Hyper-YOLO-N. Assume that a corresponding adversarial attack procedure has been applied to produce an adversarial image x^{adv} for the image x. Therefore, adversarial images are considered as inputs for evaluation and no new adversarial images are generated during evaluation for the victim detectors.

Let us define

$$\mathcal{V} = \{f_1, f_2, f_3, f_4, f_5\}$$

as a set of victim detectors that have not seen an adversarial image previously, namely, Faster R-CNN ResNet-50 FPN V2, RT-DETRv2-S, D-FINE-S, YOLOv12-S, and Deformable DETR-R50. For each $f_j \in \mathcal{V}$, both x and x^{adv} are evaluated using the same inference configuration.

The goal is to investigate if the adversarial effect generated in the source-detector setting can be observed after the adversarial image is transferred to another victim detector. There are three kinds of effects that will be investigated: target-level detection failure, changes in prediction confidence, and changes in spatial localization. Therefore, the concept of transferability cannot be understood only as the disappearance of the target in the detector output but also as the presence of the target with altered confidence or localization.

Denote by p a prediction of an object detector and by g a ground-truth target. A prediction p is a valid detection for a target g if

$$\mathrm{IoU}(p, g) \geq 0.50.$$

The ground-truth target g is considered as a candidate for adversarial testing only if it was a valid detection in the corresponding clean image. Such an approach does not allow considering targets which are missed by a detector in a clean setting as a consequence of an adversarial patch.

A target is considered successfully attacked if it can be validly detected in the clean image but does not have any valid corresponding detection in the adversarial image. The attack success rate (ASR) is calculated as

$$ASR = \frac{N_{\text{success}}}{N_{\text{eligible}}} \times 100\%.$$

If the target is validly detected in both the clean and adversarial images, the change in confidence score and localization is evaluated through the paired predictions. Suppose that c and c^{adv} represent the confidence scores of the valid paired clean and adversarial detections. In that case, the confidence change is given by

$$\Delta c = c - c^{adv}.$$

Similarly, the localization change is given by

$$\Delta IoU = IoU - IoU^{adv},$$

for the paired IoU values of IoU and IoU^{adv}. The paired values are only computed if there are valid detections in both cases. If a target disappears from detection in the adversarial image, it is considered for the attack-success analysis, but it does not receive an artificial confidence/IoU of zero.

B. Threat Model

The threat model is formulated as a transfer-based attack scenario in which the attacker has access to the adversarial images constructed in the framework of a source detector but does not craft the adversarial image against each victim detector. The Hyper-YOLO-N is considered the source detector, and the five target detectors are assumed to be unseen victim architectures.

The adversarial image sets used in this paper are fixed before testing the effect of adversarial images on the victim detectors. In addition, no new adversarial patch is created for a particular victim detector, and the current adversarial images

are not changed during the transfer attack. The evaluation of the victim detectors is conducted based on the corresponding pretrained models without any adversarial fine-tuning.

The attack objective is considered to be an evasion objective concerning the target object. Namely, the question whether the adversarial image causes the detection failure for the target object detected in the clean image is studied in this paper. Partial target detection degradation is also taken into account in this study.

The threat model can therefore be summarized as follows:

- **Source detector:** Hyper-YOLO-N, used to generate the existing adversarial samples.
- **Victim detectors:** Faster R-CNN ResNet-50 FPN V2, RT-DETRv2-S, D-FINE-S, YOLOv12-S, and Deformable DETR-R50.
- **Adversarial input:** Existing adversarial samples, alongside their respective clean samples.
- **Objects targeted:** Detection of person and cars for the transfer attack.
- **Transfer constraint:** The same adversarial samples are tested on different victim detectors without any generation, alteration of the patch, or fine-tuning of victim detectors.
- **Confidence threshold:** Detection of objects is performed at a confidence threshold of 0.25 during inference.
- **Matching criterion of target:** For a detection of a target to be valid, its IoU must be more than 0.50 with the ground truth.

Using this definition, the study measures the extent to which an adversarial influence created in a source-detector pair can be transferred to other detectors with different structures that have not been seen before. This definition also makes it possible to differentiate between total failure of detection and mere changes in confidence and localization.

IV. Methodology

The approach used in this work is designed for testing the impact of adversarial patches from the point of attack transferability as well as mitigation. The general process includes five major steps: preparing fixed adversarial image pairs, source and victim detectors evaluation, target matching on the basis of ground truth, measuring transferability quantitatively, and diffusion-based mitigation analysis. The whole process is conducted following the evaluation criteria defined in the threat model and, therefore, any differences are caused by adversarial input itself and not by target matching or selection.

A. Overall Evaluation Framework

Let x represent a clean image and x^{adv} its corresponding adversarial version. The overall experimental pipeline can be expressed as

$$\begin{aligned} x \rightarrow x^{adv} &\rightarrow \{f_1, \ldots, f_5\} \\ &\rightarrow \text{Target Matching} \rightarrow \text{Transfer Analysis} \\ &\rightarrow \text{Mitigation Analysis}. \end{aligned} \tag{1}$$

The adversarial image is created before the evaluation process of the victims and stays constant through the evaluation process of the detectors. The clean and the adversarial images for each victim detector are processed identically during the inference and target-matching processes. Thus, there will be a comparison for the reactions of the victim detector for the clean and adversarial image.

It is important to separate the failure to detect the target and the degradation of the prediction of the victim at the level of the target. If the target is no longer detected after the patching process, it will be considered an attack success. However, if the target is successfully detected but the localization and confidence scores change, the target can still be considered a failure.

B. Preparation of Adversarial Image Pairs

In this particular case, the adversarial patches that were used were already generated in the process of creating attacks. The five versions of the same diffusion-generated patches were created for the five sets of attacks and then used on their respective images. After applying the patches to the clean images, the two were kept together to form the paired samples of the evaluation process. These adversarial images were kept unaltered during the transfer evaluation phase, and no additional patch was created for any victim detectors.

The five preconstructed attack sets are denoted as

$$\mathcal{A} = \{A_{01}, A_{02}, A_{03}, A_{04}, A_{05}\}.$$

Attack sets consist of 200, 250, 300, 350, and 400 images, providing a total of 1,500 clean-adversarial image pairs. All the attack sets include equally divided classes of person and car target images. Corresponding patches for diffusion generation belong to their respective attack sets and are used in them consistently.

The location of the patch is chosen in relation to the ground-truth of the selected target image. Patch size is set to 30% of the smaller dimension of the ground-truth bounding box of the target image, and it is applied to the center of the bounding box. In the process of patch application, its rotation equals 0°, brightness does not change, and its opacity equals 100%. Such consistency in the chosen parameters ensures that the effect of the same preconstructed adversarial inputs on diverse victim detector architectures is studied.

For each of the images, the clean and adversarial counterparts have the same background and target image, as well as the same ground-truth of the latter. Hence, the adversarial patch is the variation between the two images in this setup.

C. Source-Detector and Victim-Detector Processing

Hyper-YOLO-N is treated as the source-detector context from which the existing adversarial images originate. The transfer evaluation is then performed on five unseen victim detector architectures:

$$\mathcal{V} = \{f_1, f_2, f_3, f_4, f_5\},$$

where the five victim detectors are Faster R-CNN ResNet-50 FPN V2, RT-DETRv2-S, D-FINE-S, YOLOv12-S, and Deformable DETR-R50.

For every victim detector $f_j \in \mathcal{V}$, both the clean and adversarial versions of the input x and x^{adv}, respectively, are analyzed by the detector separately. The same confidence threshold is used for both scenarios in order to avoid confusion arising from any differences in detection filtering.

The transfer procedure does not involve adversarial training of the victim detectors. Namely, neither of the victim detectors is fine-tuned using the adversarial images, nor is any adversarial patch optimized in order to increase its effectiveness for the particular victim detector.

D. Ground-Truth-Based Target Matching

A ground-truth-based matching procedure is used to see if the detector can successfully detect the selected target. Let g be the chosen ground-truth bounding box and let p_i be the predicted bounding box. A prediction qualifies as a target detection only if it meets the following two requirements:

$$\text{class}(p_i) = \text{class}(g)$$

and

$$IoU(p_i, g) \geq 0.50.$$

For each target in the ground truth, only those predictions which belong to the target class are taken into consideration. In case there are multiple predictions of the same class, then only the prediction with the maximum IoU with respect to the chosen ground-truth bounding box is used for evaluation at the target level.

This process is carried out separately in both the clean and adversarial cases. The use of the same chosen ground-truth target in both cases helps ensure that the change in the response of the detector is observed for the same object rather than for any other object.

E. Target Eligibility and Attack Evaluation

The incorporation of a target into adversarial evaluation depends on its valid detection in a clean image. The target is therefore defined as an eligible target in the following manner:

$$E(g) = \begin{cases} 1, & \text{if there is a valid clean detection,} \\ 0, & \text{otherwise.} \end{cases}$$

The above-mentioned eligibility requirement is crucial since any target not detected by the detector in the clean image cannot form an example of a successful adversarial suppression event.

In order to assess whether an eligible target was successfully attacked, the result of adversarial detection is compared to the clean detection result for each eligible target. Attack-success events are characterized as follows:

$$D_{\text{clean}}(g) = 1 \quad \text{and} \quad D_{\text{adv}}(g) = 0,$$

where D refers to the presence of a valid target detection.

Attack Success Rate (ASR) is calculated according to the formula:

$$ASR = \frac{N_{\text{success}}}{N_{\text{eligible}}} \times 100\%.$$

Therefore, the denominator equals the number of valid clean detections and does not refer to the number of input images. Clean misses are not included in the attack-success denominator since they cannot be attributed to an adversarial patch.

F. Confidence and Localization Degradation

Complete target disappearance is not sufficient to quantify all effects of an adversarial patch. If a target is validly detected in both clean and adversarial settings, the comparison between clean and adversarial detections is performed using the confidence and localization metrics of the matched targets.

Denote by c the confidence of the clean detection and c^{adv} the confidence of the adversarial one. The confidence change is then defined as

$$\Delta c = c - c^{adv}.$$

The value of Δc is positive in the case of a decrease in confidence due to the patch and negative in the case of an increase.

The localization change is quantified in a similar way. Denote by IoU the IoU of the clean detection with respect to the chosen ground-truth target and IoU^{adv} the IoU of the adversarial one. The localization change is then defined as

$$\Delta IoU = IoU - IoU^{adv}.$$

The values Δc and ΔIoU are calculated only for targets that can be detected in both settings. In the case of a successful suppression, the target contributes to ASR, but no artificial confidence or IoU is assigned to it. This helps to avoid misclassifying a missed target as a zero-confidence and zero-localization detection.

For each attack set and victim detector, the following information is gathered: the number of eligible targets, number of successful attacks, ASR, paired confidence changes, paired IoU changes, and number of targets that have both clean and adversarial detections. Class-specific information is kept separately for the person and car target classes.

G. Cross-Architecture Transfer Evaluation

The transferability experiment performs the evaluation of each fixed adversarial set of images separately for all five victim detector architectures. Let $R_{k,j}$ be the result of the evaluation for the attack set A_k on victim detector f_j, where

$$k \in \{1, 2, 3, 4, 5\} \quad \text{and} \quad j \in \{1, 2, 3, 4, 5\}.$$

Therefore, each attack set is tested on all five victim detectors in turn without any modifications of the adversarial input. This results in the 5×5 experimental design, allowing for a direct comparison of the same attack family across different detector architectures.

Transferability analysis considers three different levels of adversarial effect. First, it includes target suppression, measured by the Attack Success Rate (ASR). Second, it takes into account prediction-confidence degradation, Δc. Third, it evaluates localization degradation, ΔIoU. The multi-level approach allows distinguishing between those adversarial attacks that eliminate a target object from the detector output and those that keep it detectable but impair the predictive performance of the detector.

Negative Δc and/or ΔIoU values are preserved without clipping to zero because the actual paired detector responses are considered.

H. Diffusion-Based Mitigation

In addition to the study of transferability, the current paper explores diffusion-based adversarial patch mitigation. Adversarial patch mitigation follows the approach proposed in the framework of DisPatch, wherein image regeneration is used for identifying anomalous regions followed by selective image restoration [7].

The process of adversarial patch mitigation involves two phases. First, the image that is subject to attack is regenerated using diffusion-based image inpainting. Image regeneration uses spatial complementing to reconstruct the regions in the image while keeping the context of the rest of the image intact. Let us denote the regenerated image of the attacked image x^{adv} by $\tilde{x}$.

The original and regenerated images are then compared pixel-wise to obtain an image-difference map:

$$D(u,v) = \left\| x^{adv}(u,v) - \tilde{x}(u,v) \right\|_2 .$$

The resulting difference map is then smoothed and segmented into areas of low and high differences. Next, the two-cluster K-means clustering method is applied to separate regions with minor changes in the reconstructed images from those with major differences. The high-difference area is considered as the prediction of the adversarial region.

Afterwards, the adversarial area is selectively substituted with the reconstruction generated by the diffusion process, whereas the rest of the image is retained as it is in the original adversarial image. This allows us to represent the corrected image as follows:

$$x^{rect} = A \odot \tilde{x} + (1 - A) \odot x^{adv},$$

where A is the adversarial region mask and $\odot$ is element-wise multiplication.

It is important that only the adversarial area is substituted by the reconstructed image rather than the whole input image since the use of the reconstruction of the whole image may change the benign content of the image.

I. Known-Location Restoration Diagnostic

To test the relative effects of diffusion restoration quality and adversarial region localization accuracy, the additional diagnostic experiment with known location is performed independently from the automatic detection pipeline.

In this experimental setup, the real patch mask is directly fed to the diffusion inpainting module. As a result, the automatic localization steps, i.e., the calculation of the difference between the original and regenerated images and the K-means-based mask prediction, are skipped. The experiment is designed to check if the diffusion network is able to restore the damaged patch given the accurate knowledge of its position.

The experiment is characterized by the following transformation:

$$x^{known} = A_{GT} \odot \tilde{x} + (1 - A_{GT}) \odot x^{adv},$$

where A_{GT} is the ground-truth patch mask. It is possible to include one more enlarged patch mask into consideration in order to find out if the increase in the restoration region helps to smooth boundaries.

The initial image is used only for evaluation purposes; its pixels are not available to the restoration process. Such restriction rules out the possibility of oracle reconstruction of the damaged image based on the copying of hidden pristine pixels.

In other words, the known-location experiment is taken as a diagnostic test rather than an estimation of the automatic detection method's performance. Successful restoration in the known-location case may indicate the inefficiency of localization, while continued failure may indicate a limitation of diffusion or detector-based restoration.

J. Preservation and Restoration Evaluation

The mitigation stage is evaluated at both the image and detector levels. For attack-success cases, a restoration is considered successful when a target that was validly detected in the clean condition, suppressed in the attacked condition, and subsequently evaluated after mitigation becomes validly detected again.

For control cases that are not initially suppressed by the attack, the analysis checks whether the mitigation preserves their existing valid detection. This produces two complementary concepts: target recovery and control preservation.

At the image level, the study also examines whether changes outside the intended restoration region remain negligible. In the known-location diagnostic, the outside-mask difference between the attacked input and the final restored image is measured to verify that the restoration procedure changes only the designated region. Reconstruction similarity inside the true patch area can additionally be evaluated using pixel-level measures such as mean absolute error (MAE), mean squared error (MSE), peak signal-to-noise ratio (PSNR), and structural similarity (SSIM), where the image dimensions permit valid computation. These image-level measures are used only for diagnosis and do not participate in the restoration process.

V. Experimental Setup

In this part, the dataset used, the attack samples, the detection parameters, the inference environment, and the mitigation conditions are specified to conduct the experiments. All experiments conform to the conditions of target matching

and eligible criteria set out in Section III and the procedure of processing mentioned in Section IV. For the transfer experiments, there are always predefined adversarial image pairs for the consistency of experiments.

A. Dataset and Target Selection

The image sets are from the MS COCO 2017 validation set. The clean-image test has been conducted using the first 500 images of the validation set by ordering the filenames to ensure the baseline performance of the pretrained Hyper-YOLO-N detector. For this test, no fine-tuning or additional training has been performed on the pretrained detector, with an input size of 640×640 and a confidence threshold of 0.25.

The main adversarial transfer testing uses five previously created attack sets from the attack pipeline of the project. Testing is limited to only two target classes, namely *person* and *car*. Both target classes have an equal number of images in each attack set. The sizes of these attack sets are shown in Table I.

TABLE I: Composition of the five fixed adversarial attack sets.

Attack Set	Total	Person	Car
A_{01}	200	100	100
A_{02}	250	125	125
A_{03}	300	150	150
A_{04}	350	175	175
A_{05}	400	200	200
Total	**1500**	**750**	**750**

There are 1,500 pairs of clean images and adversarial examples from the five attack sets. Every clean image and corresponding adversarial image pair remain together throughout the entire experiment. The source code confirms that the two target classes, the five increasing sizes of the attack sets, and the patch attack application method have been used. [10]

B. Preconstructed Adversarial Inputs

The patches used in the attacks were designed prior to the transfer evaluation and were not optimized for the victim detectors. There were five different patches, which were used for each of the five attack sets. After being generated, each of the patches was applied to the corresponding target images, forming attacked image sets. These attacked images were kept constant throughout the transfer process.

In the source attack, the position of the patch in the attacked image is related to the target ground truth. The size of the patch is set to 30% of the shorter side of the bounding box of the ground-truth target. The patch is placed at the center of the bounding box, rotated by 0°, without any change in brightness and with full opacity. The diffusion algorithm used in the project to generate candidate patches is Stable Diffusion 2.1 Base.

Neither patch was altered or re-optimized during the evaluation of the fixed attacked images on the victim detectors. Thus, the transfer benchmark evaluates the behavior of preexisting adversarial inputs on different detector architectures, rather than the ability to generate new attack algorithms for each detector separately.

C. Source and Victim Detectors

Hyper-YOLO-N represents the source-detector context from which all of the adversarial images used in the present experiment are derived. Hyper-YOLO is an extension of YOLO via hypergraph-based feature computation and is thus used as the source detector for transfer evaluation [10].

The transfer learning benchmark consists of five unseen victim detectors. Faster R-CNN with ResNet-50 FPN V2 represents a two-stage region-based detector [12]. RT-DETRv2-S is a real-time, transformer-based detector [14]. D-FINE-S represents fine-grained distribution refinement for bounding-box regression [15]. YOLOv12-S is an attention-centric YOLO detector [16]. Deformable DETR-R50 is a deformable-attention transformer detector [13].

All five detectors mentioned above are tested with their pretrained weights, and none of them undergo adversarial fine-tuning or attack-specific optimization. The use of heterogeneous detectors is intended to find out if the adversarial effect generated by Hyper-YOLO can be observed on detectors having different architectures.

D. Inference Configuration

The Hyper-YOLO experiments use an input resolution of 640×640 with a confidence threshold value of 0.25 and an NMS IoU threshold of 0.70. The definition of the ground-truth target match for the source attack is at an IoU threshold value of 0.50.

In the victim-detector transfer experiment setup, the confidence threshold value is kept constant at 0.25 for all detector evaluations. The ground-truth matching threshold is kept constant at 0.50 for finding valid targets. The clean and adversarial images undergo processing separately according to the detector-specific inference setup without differences in confidence filtering between the two conditions.

The requirement for a valid predicted target is class matching and achieving an IoU score of 0.50 or more with the chosen ground-truth target. The target-level validation selects the predicted target with the highest IoU among the same-class predictions.

E. Transfer Evaluation Protocol

Each of the five preset attack sets is individually evaluated on all five victim detectors, resulting in a 5×5 cross-architecture evaluation matrix, whereby all attacks are applied to all victim architectures.

The evaluation starts with identifying valid detections on clean images for each detector and attack set. These clean-valid detections are considered eligible for further evaluation of attack success rates; a clean miss is not included in the denominator of ASR.

For each valid detection, an evaluation is performed on the corresponding attacked image and the same ground-truth target. If the target is still validly detected after patching, the

confidence and IoU values are recorded. If not, it is considered an attack-success case.

Changes in confidence and IoU are measured only for those cases when both the original and attacked detections are valid. Missing detections are not replaced with zero, so the difference between complete removal of the target from the detections and its partial degradation can be evaluated.

The evaluation is done separately for *person* and *car* classes in addition to the overall results. The purpose of this class-wise evaluation is to investigate the difference in transferability between the selected classes.

F. Baseline and Control Conditions

The preliminary test of Hyper-YOLO on clean images provides a baseline for the source detector before the adversarial test is conducted. This preliminary test was done using 500 COCO validation images, where prediction-level statistics were obtained without replacing the official detection metrics based on ground truth. The objective was to ensure that the source detector was behaving as expected before the adversarial transfer test was performed.

The control images are part of the mitigation test together with the successfully attacked images. Control images refer to those images where after the adversarial attack, the target is still successfully detected. The objective of including controls is to determine whether the mitigation retains a correct detection or not.

The use of paired conditions (clean, attacked, and control) makes it possible to differentiate between adversarial and detector behaviors in clean images.

G. Diffusion-Based Mitigation Configuration

For the mitigation analysis, DisPatch is chosen as the reference defense framework that works on a diffusion-based approach (Ma et al., 2025). The official automatic mitigation experiment is based on a set of 30 images that is taken from the Hyper-YOLO attack benchmark, including 15 attack-success and 15 control cases.

Automatic mitigation exploits the CompVis latent diffusion inpainting model with the configuration called `inpainting_big`. Diffusion regeneration is performed using the DDIM sampler with five sampling steps. The checkerboard regeneration configuration uses $N = 32$. Automatic mitigation is performed using the fixed seed 20260817.

The automatic defense does not have knowledge of the actual patch coordinates. The attacked image is regenerated in accordance with the checkerboard configuration, and a difference map between the original and regenerated images is calculated, which allows the suspicious-region mask to be obtained automatically for further selective restoration of the attacked image.

Pixels of the clean image are not used for the generation of the recovered image. Therefore, the question is whether diffusion-based regeneration and automatic localization allow the detector response to be recovered without adding the clean target patch to the attacked image.

H. Known-Location Restoration Diagnostic

An independent diagnostic evaluation is performed for the known-location scenario to differentiate between the problem of adversarial-region localization and the diffusion model's ability to reconstruct the damaged area. In the diagnostics, only five images (including four attack-success cases and one control case) are used.

When the known location scenario is applied, the exact coordinates of the true patch are provided directly using the binary restoration mask. The automatic difference map and K-means localization stages are skipped. There are two types of masks that are used for the restoration: exact patch mask and the mask with the increased boundary by 4 pixels around the patch.

The same LDM-based CompVis inpainting pipeline is utilized as before, with DDIM running during five sampling steps. The clean image is used only for evaluation and visualization of the results after restoration, but not for input data to the restoration stage. The result of the restoration procedure is created by replacing the restoration area with images generated from the diffusion procedure, while the attacked image is kept untouched in other areas.

I. Evaluation of Mitigation Outputs

Mitigation effectiveness is evaluated at the detector and image levels. In terms of the detector level, a target attack success case can be said to have been recovered if the target is detected successfully in the uncorrupted state, rendered invalid by the attack, and then successfully detected again post-mitigation. Control maintenance is separately evaluated for those targets which still remain valid after the attack.

In the image level evaluation, the automatic mitigation performance is evaluated in terms of its prediction accuracy of adversarial regions. In terms of the known location test, outside-mask integrity is verified to make sure that those pixels located outside of the targeted restoration region are left untouched. The reconstruction similarity within the true patch region can also be measured in terms of the difference in comparison to the clean image.

There is a reason for keeping these two evaluations separate from one another as image restoration does not necessarily coincide with detector recovery and vice versa. Restoration of an image can change the look of the attacked region without restoring the target detection, while detector can restore the target detection without restoring the region to its original pixel state.

J. Implementation and Reproducibility

The experiments are conducted using the available implementations of the Hyper-YOLO, adversarial-attack, and DISPATCH projects. The original Hyper-YOLO attack implementation code, as well as the frozen attack artifacts, are used as read-only inputs in the transfer benchmark. The detection and mitigation outputs of all victim detectors are stored independently of the attack inputs.

The experiments are conducted in a Windows environment with an NVIDIA RTX 2050 GPU. In the case of the Deformable DETR-R50 experiment, the pretrained weights from SenseTime were used due to the unavailability of the `ms_deform_attn` CUDA implementation provided by the official implementation on the Windows host. The implementation used in this experiment was accessed through the Hugging Face Transformers library. This fact is reported to ensure reproducibility and to distinguish the constraint in the detector implementation from the attack evaluation procedure.

The following principles are upheld throughout the experiments: fixed clean–adversarial image pairs, identical adversarial images in the transfer test, identical target classes, an identical confidence threshold value, identical ground-truth matching criteria, no victim-specific adversarial optimization, no victim-specific adversarial fine-tuning, and isolation of automatic mitigation and known-location diagnostic output storage.

VI. Experimental Results

In this section, we show the empirical findings for the adversarial patch transfer experiment and the mitigation experiment using diffusion-based defense. The discussion starts with the general transfer performance over the five victim detectors that were not seen during training and continues by considering attack-wise variations, per-class transfer performance, confidence drops, localization differences, and diffusion-based mitigation. Transfer performance follows the rules described in Sections III and IV.

A. Overall Transfer Results

In the finished benchmark for transfer attacks, five attack sets, namely A_{01} to A_{05}, are used on five different victim detectors. The five victim detectors are tested with the same 1,500 pairs of images (clean and adversarial images), which means that there are 7,500 detector-image combinations for the whole benchmark.

As the criteria for eligible targets are determined separately for each victim detector with respect to its predictions on clean images, the number of eligible targets is different for each detector. In total, there are 6,653 eligible target instances and 168 successful attacks among the five victim detectors, which means that the attack success rate is 2.53%.

Table II summarizes the aggregate results for the five victim detectors.

TABLE II: Aggregate adversarial transfer results across the five unseen victim detectors.

Detector	Total	Eligible	Success	ASR	Person	Car
Faster R-CNN	1500	1365	16	1.17%	0.45%	1.87%
RT-DETRv2-S	1500	1345	32	2.38%	2.40%	2.36%
D-FINE-S	1500	1367	23	1.68%	1.34%	2.01%
YOLOv12-S	1500	1267	49	3.87%	1.54%	6.31%
Deformable DETR-R50	1500	1309	48	3.67%	2.43%	4.92%

The observed aggregate ASR varies across the five architectures. Faster R-CNN records 1.17%, RT-DETRv2-S records 2.38%, D-FINE-S records 1.68%, YOLOv12-S records 3.87%, and Deformable DETR-R50 records 3.67%. These values represent observed transfer effects under the experimental protocol and are not detector-quality scores.

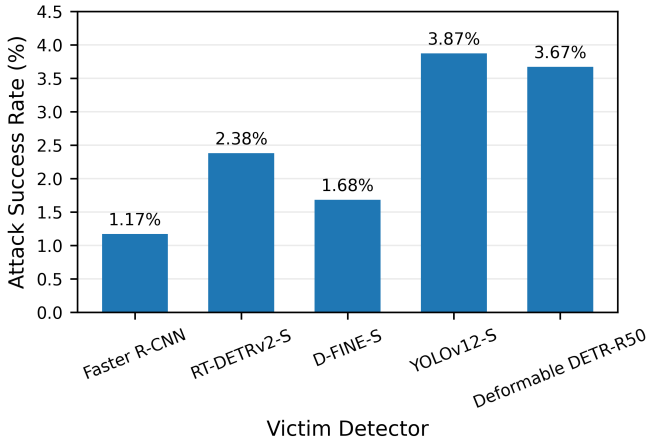


Fig. 1: Overall attack success rate across the five unseen victim detectors.

Figure 1 provides a visual representation of the aggregate ASR values and complements the numerical results in Table II.

B. Attack-Level Transfer Results

The five fixed attack sets contain 200, 250, 300, 350, and 400 images, respectively. Their corresponding transfer results are evaluated independently on each victim detector.

Table III presents the complete attack-level transfer results. "Both-valid" denotes the number of eligible targets for which valid detections remained available in both the clean and adversarial conditions.

The attack-level results demonstrate that transfer behavior changes across both detector architectures and attack sets. Faster R-CNN records its highest ASR in A_{05} at 1.93%, while RT-DETRv2-S reaches 2.88% in A_{04}. D-FINE-S reaches 2.62% in A_{02}, YOLOv12-S reaches 5.39% in A_{04}, and Deformable DETR-R50 reaches 3.80% in A_{03}.

TABLE III: Attack-level transfer results for all five victim detectors. Confidence and IoU changes are calculated only on both-valid cases.

Detector	Attack	Total	Eligible	Success	ASR	Δc	ΔIoU
Faster R-CNN	A01	200	187	2	1.07%	0.006666	0.003081
Faster R-CNN	A02	250	225	2	0.89%	0.009843	0.001531
Faster R-CNN	A03	300	277	1	0.36%	0.007127	-0.001163
Faster R-CNN	A04	350	314	4	1.27%	0.020942	0.001895
Faster R-CNN	A05	400	362	7	1.93%	0.013950	0.001469
RT-DETRv2-S	A01	200	184	3	1.63%	0.013282	0.005105
RT-DETRv2-S	A02	250	226	6	2.65%	0.041074	0.008820
RT-DETRv2-S	A03	300	264	4	1.52%	0.013776	0.005740
RT-DETRv2-S	A04	350	312	9	2.88%	0.018818	0.008750
RT-DETRv2-S	A05	400	359	10	2.79%	0.025238	0.003591
D-FINE-S	A01	200	185	1	0.54%	0.026847	0.004872
D-FINE-S	A02	250	229	6	2.62%	0.016758	0.006173
D-FINE-S	A03	300	276	6	2.17%	0.020004	0.002793
D-FINE-S	A04	350	314	5	1.59%	0.016473	0.005923
D-FINE-S	A05	400	363	5	1.38%	0.021184	0.003974
YOLOv12-S	A01	200	172	7	4.07%	0.021915	0.009103
YOLOv12-S	A02	250	207	8	3.86%	0.027582	-0.001438
YOLOv12-S	A03	300	250	3	1.20%	0.025300	-0.000169
YOLOv12-S	A04	350	297	16	5.39%	0.016757	0.000630
YOLOv12-S	A05	400	341	15	4.40%	0.031621	0.004965
Deformable DETR-R50	A01	200	180	6	3.33%	0.028476	0.002795
Deformable DETR-R50	A02	250	216	8	3.70%	0.015151	0.004970
Deformable DETR-R50	A03	300	263	10	3.80%	0.006091	0.002385
Deformable DETR-R50	A04	350	304	11	3.62%	0.026969	0.003450
Deformable DETR-R50	A05	400	346	13	3.76%	0.024734	0.003044

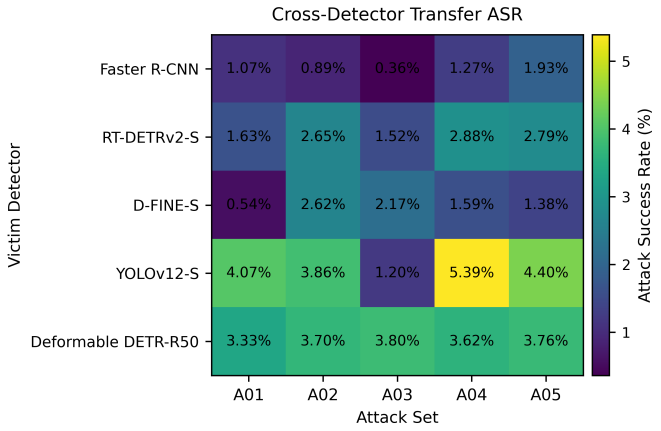


Fig. 2: Cross-detector transfer heatmap showing ASR for attack sets A_{01}–A_{05}.

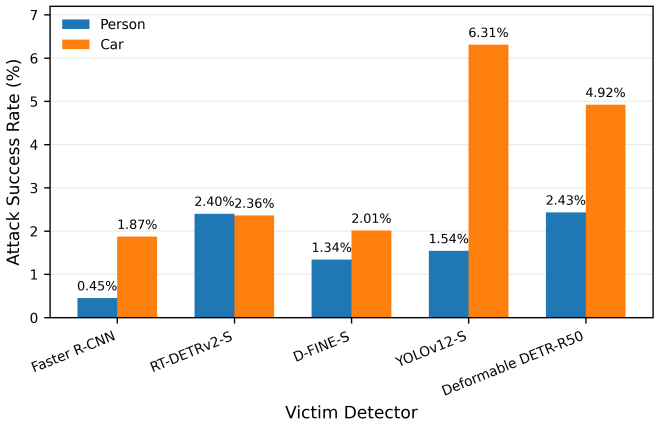


Fig. 3: Person-versus-car attack success rates across the five victim detectors.

Figure 2 presents the complete 5×5 detector–attack transfer matrix and illustrates the variation in observed ASR across the fixed attack sets and unseen victim architectures.

C. Class-Specific Transfer Results

The transfer benchmark separately evaluates person and car targets. Table IV reports the eligible targets, successful attacks, and ASR values for both classes.

The aggregate class-specific results show 0.45% person ASR and 1.87% car ASR for Faster R-CNN, 2.40% and 2.36% for RT-DETRv2-S, 1.34% and 2.01% for D-FINE-S, 1.54% and 6.31% for YOLOv12-S, and 2.43% and 4.92% for Deformable DETR-R50.

For Faster R-CNN, D-FINE-S, YOLOv12-S, and Deformable DETR-R50, the aggregate car ASR is higher than the corresponding person ASR. For RT-DETRv2-S, the two class-specific values are closely matched.

D. Confidence Degradation

The attack success rate encompasses target validity loss, but an adversarial patch may decrease target classification confidence while still making the target valid. Confidence

TABLE IV: Class-specific transfer results. Each entry reports Eligible / Success / ASR.

Attack	Faster R-CNN	RT-DETRv2-S	D-FINE-S	YOLOv12-S	Deformable DETR-R50
A01 Person	92/1/1.09%	91/2/2.20%	91/0/0.00%	89/1/1.12%	91/3/3.30%
A01 Car	95/1/1.05%	93/1/1.08%	94/1/1.06%	83/6/7.23%	89/3/3.37%
A02 Person	113/1/0.88%	115/4/3.48%	114/2/1.75%	109/4/3.67%	109/2/1.83%
A02 Car	112/1/0.89%	111/2/1.80%	115/4/3.48%	98/4/4.08%	107/6/5.61%
A03 Person	134/0/0.00%	130/2/1.54%	135/1/0.74%	128/0/0.00%	133/3/2.26%
A03 Car	143/1/0.70%	134/2/1.49%	141/5/3.55%	122/3/2.46%	130/7/5.38%
A04 Person	151/1/0.66%	152/4/2.63%	153/4/2.61%	148/2/1.35%	151/3/1.99%
A04 Car	163/3/1.84%	160/5/3.13%	161/1/0.62%	149/14/9.40%	153/8/5.23%
A05 Person	178/0/0.00%	178/4/2.25%	179/2/1.12%	175/3/1.71%	175/5/2.86%
A05 Car	184/7/3.80%	181/6/3.31%	184/3/1.63%	166/12/7.23%	171/8/4.68%

change is then calculated only for those targets which are valid in both the clean image and the adversarial case.

The weighted mean confidence change is 0.012483 for Faster R-CNN, 0.022492 for RT-DETRv2-S, 0.019905 for D-FINE-S, 0.024935 for YOLOv12-S, and 0.020449 for Deformable DETR-R50.

TABLE V: Aggregate confidence degradation for both-valid target matches.

Detector	Both-valid	Δc	Median
Faster R-CNN	1349	0.012483	0.000140
RT-DETRv2-S	1313	0.022492	0.002234
D-FINE-S	1344	0.019905	0.003131
YOLOv12-S	1218	0.024935	0.004864
Deformable DETR-R50	1261	0.020449	0.006432

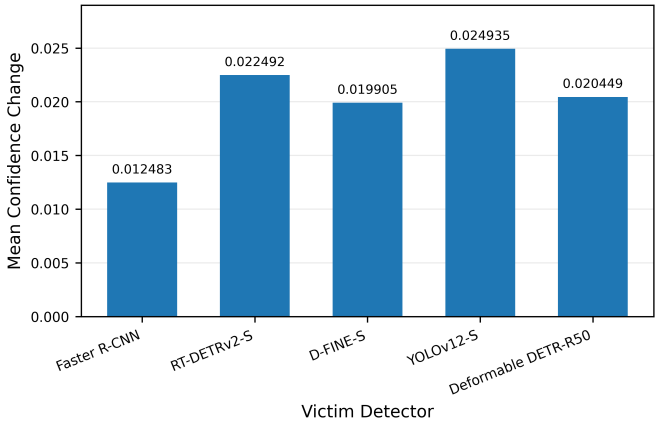


Fig. 4: Weighted mean confidence change for both-valid target matches.

Figure 4 shows that confidence degradation remains measurable even when the target is not completely suppressed. Therefore, ASR alone does not capture all observed effects of the adversarial images.

E. Localization Degradation

Localization modifications are obtained by calculating the discrepancy between the IoU values of the clean image and adversarial image. The IoU changes for the weighted mean are as follows: 0.001260 for Faster R-CNN, 0.006292 for RT-DETRv2-S, 0.004673 for D-FINE-S, 0.002438 for YOLOv12-S, and 0.003289 for Deformable DETR-R50.

TABLE VI: Aggregate localization degradation for both-valid target matches.

Detector	Both-valid	ΔIoU	Median
Faster R-CNN	1349	0.001260	0.001531
RT-DETRv2-S	1313	0.006292	0.005740
D-FINE-S	1344	0.004673	0.004872
YOLOv12-S	1218	0.002438	0.000630
Deformable DETR-R50	1261	0.003289	0.003044

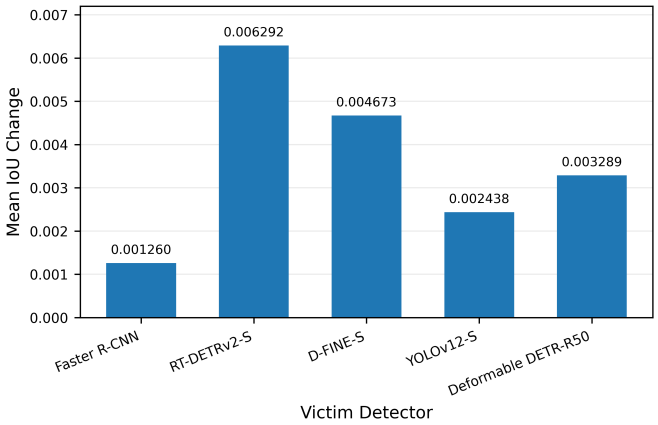


Fig. 5: Weighted mean IoU change for both-valid target matches.

The attack-level results also show that there is no consistency in terms of the direction of the localization change. Negative mean IoU changes have been recorded for Faster R-CNN on A_{03} and YOLOv12-S on A_{02} and A_{03}. These values have not been altered since they represent the true measure of the paired responses.

F. Cross-Detector Transfer Behavior

The results of the cross-detector attacks show that the same fixed adversarial inputs generate different target-level impacts on different detector architectures. For example, the ASR values are 1.17%, 2.38%, 1.68%, 3.87%, and 3.67% for

Faster R-CNN, RT-DETRv2-S, D-FINE-S, YOLOv12-S, and Deformable DETR-R50, respectively.

The transfer effect is different not only among different detector architectures but also among the five attack sets. Thus, ASR, confidence degradation, and localization degradation provide different yet complementary measures of the impact of an attack.

The number of eligible targets differs among the five detectors. This stems from the definition of clean-image eligibility since only those targets which can be correctly detected under the clean condition are considered in the ASR denominator.

G. Automatic Diffusion-Based Mitigation

The mitigation experiment aims at examining if regeneration based on diffusion techniques and suspicious-region localization can be successful in recovering the objects that are suppressed due to the adversarial patch. The official automatic experiment is called `mitigation_01` and works with 30 images obtained from the Hyper-YOLO attack benchmark. It contains 15 attack-success examples and 15 control cases.

All 30 images have been processed successfully. Two out of 15 attack-success targets have been recovered, which gives a DRR of 13.33%. Thirteen out of 15 control cases have been kept intact, which results in a CPR of 86.67%, and two control cases regressed.

TABLE VII: Official automatic diffusion-based mitigation results.

Metric	Result
Selected images	30
Processed images	30
Attack-success cases	15
Recovered targets	2
Detection Recovery Rate	13.33%
Remaining failed targets	13
Control cases	15
Controls preserved	13
Control Preservation Rate	86.67%
Control regression	2
Person recovery	0/4
Car recovery	2/11
Mean confidence recovery	-0.098558
Mean IoU recovery	-0.017907
Mean mask IoU	0.069325
Mask precision	0.090624
Mask recall	0.775688
Mean processing time	114.24 s/image

For the mean IoU of the adversarial mask, we have 0.069325. The mask precision and recall for the same will be 0.090624 and 0.775688, respectively. This is due to the fact that the localization measures help us gain another perspective on the automatic protection technique.

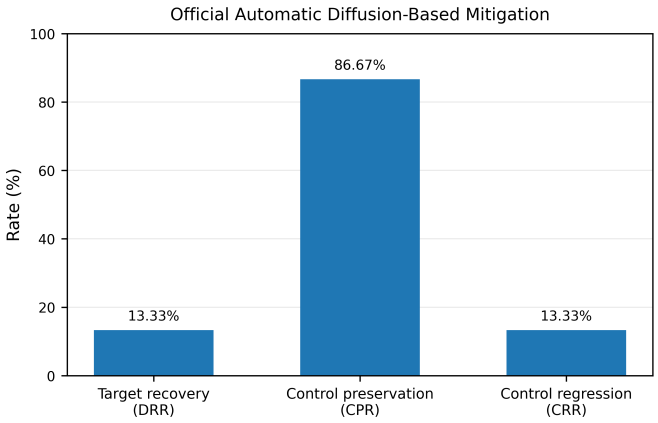


Fig. 6: Official automatic diffusion-based mitigation results showing detection recovery and control preservation.

Hence, the formal mitigation experiment led to partial restoration of the detectors while demonstrating limited correspondence between the suspicious regions auto-detected and the actual adversarial patch regions..

H. Known-Location Diffusion Restoration Diagnostic

A known-location diagnosis was independently performed to separate the impact of adversarial region localization from the restoration capability of the diffusion model.

TABLE VIII: Five-image known-location diffusion restoration diagnostic.

Condition	Recovered	Control	Outside-Mask MAE
Automatic	0/4	1/1	–
Known, exact mask	2/4	1/1	0.000000
Known, +4 px mask	2/4	1/1	0.000000

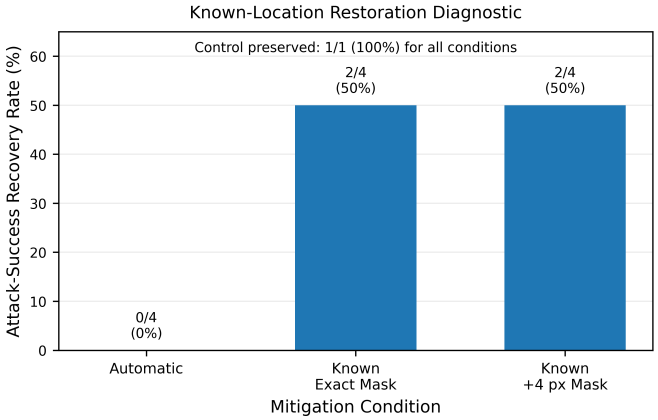


Fig. 7: Known-location diagnostic comparing automatic localization, exact-mask restoration, and $+4$ pixel mask restoration.

In the diagnosis process, five images, comprising four attack successful images and one control image, were used.

The known location condition, unlike the automatic process, is given the actual patch shape. This eliminates the need for

the automatic difference map localization stage and K-means clustering. Restoration with the exact patch mask and the expanded patch mask, by 4 pixels, was then compared.

In the diagnostic test set, the automatic localization pipeline did not recover any of the four successful attacks, while the two known-location approaches each recovered two. The outside mask MAE was 0.0 for both known-location approaches, meaning that the known-location composition technique retained pixels that fell outside the restoration zone.

From the diagnostic test, it is clear that localization can have an impact on detection recovery. But there were still two successful attacks that could not be recovered even with the right geometry of the patches.

I. Automatic Versus Known-Location Analysis

These two are completely different sets of experiments because the automatic experiment is done with the aim of finding the suspicious region without having the true patch coordinates while the known-location experiment gives the true patch coordinates of the restoration region.

The diagnostic of the five images has shown some improvement where recovery is improved from 0/4 in the automatic experiment to 2/4 in both the known location experiments. But the recovery done using the known locations is not complete. This shows that sometimes localization plays a role and in other times, restoration limits the process.

Therefore, these two experiments cannot be considered as equivalent defense strategies because the automatic experiment is the result of the automatic mitigation pathway while the known location is used to test the restoration.

J. Qualitative Restoration Analysis

Quantitative measures of detection and localization recovery fail to fully assess the visual performance of the restoration based on diffusion. Therefore, the images from the diagnostic for known-location restoration are analyzed qualitatively by comparing the clean image, the adversarial patch image, the true patch mask, and the diffusion-restored image.

For known location restoration, the true patch area, which can be determined using the information about the attack generation procedure, is directly input into the restoration stage. Thus, the automatic localization of the adversarial patch is avoided, making it possible to test the reconstruction capability of the diffusion network in the affected region. The clean image acts only as the reference image for evaluation purposes.

Qualitative evaluation is aimed at understanding whether the output obtained via the diffusion process creates a visually plausible restoration of the region under consideration. The restored image is compared to the clean image as a reference, whereas the adversarial image and the ground truth patch mask are used for contextualization to detect the changed region.

Figure 8 shows the series of images that starts from the clean input image, goes through the patched image and its ground-truth patch mask, and ends at the image after diffusion restoration. The qualitative restoration interpretation is conducted in the context of the detector metrics because it is possible for the image to look restored while the detection is not recovered, or vice versa.

K. Additional Diffusion-Based Restoration Analysis

Another restoration experiment was performed using the `stable-diffusion-2-1-base` model [20]. In this particular experiment, it was intended to see whether diffusion-based synthesis would be able to restore the damaged part caused by the adversarial patch once the location of that patch is known.

Figure 9 shows the process of restoration and the transition from the initial clean image to the patched one, to the patch mask, and to the final restored output image. Restoration makes the perceptual effect of the adversarial patch weaker and restores the corrupted region using newly generated visual content.

In the process, the attacked image and the patch mask were employed to demarcate the damaged portion of the image. Afterward, the `stable-diffusion-2-1-base` model was used to generate a restored version of that portion. In this experiment, the original image was only used for comparison and testing, but not to replace the damaged pixels.

The restored image does not have to be identical to the original one, pixel by pixel. Since the restoration is performed using a diffusion model, the synthetic content can be different from the initial content in terms of texture, visual appearance, and some details while being visually restored nonetheless. Therefore, the restoration results are considered separately from the results of the automatic DISPATCH mitigation and are shown as part of the qualitative evaluation of the defense performance.

This test provides an alternative perspective on the capability of diffusion restoration to mitigate the effects of adversarial patches.

L. Summary of Experimental Findings

All completed experiments result in three interrelated observations. First, the same fixed Hyper-YOLO adversarial images have transfer effects on five unseen detector architectures. Second, transferred adversarial effects are not limited to target suppression; rather, such effects may include prediction confidence and localization changes in detections that remain valid. Third, the diffusion-based mitigation technique partially restores targets, while automatic adversarial-region localization adds another limitation that can be quantitatively estimated with the known-location test.

In total, across all five victim detectors, there were 168 attack-success events out of 6,653 valid target instances. Attack success rate varies depending on the architecture, and class-based statistics show the difference in the behavior of person and car targets. Additional confidence and IoU measurements allow for detecting the presence of adversarial effects even in those cases where the target is not fully suppressed.

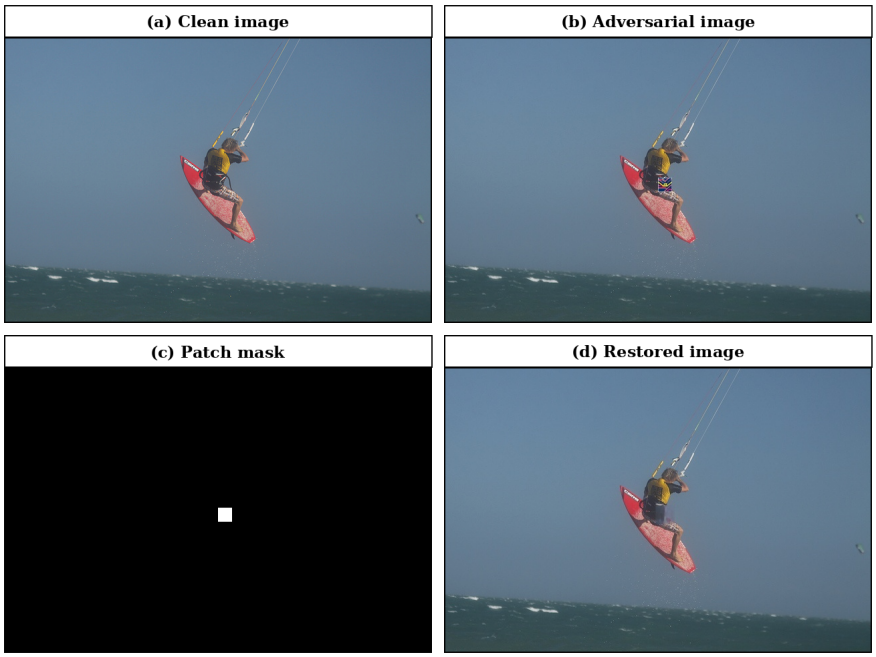


Fig. 8: Qualitative example from the known-location restoration diagnostic showing the clean image, adversarially patched image, true patch mask, and diffusion-restored image.

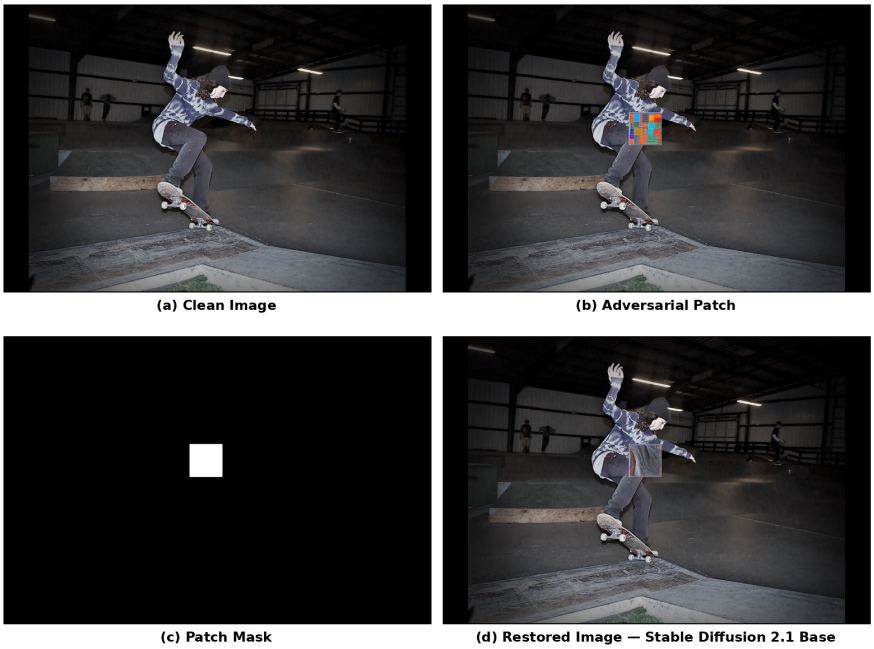


Fig. 9: Qualitative example of diffusion-based restoration showing the clean image, adversarially patched image, patch mask, and restored image generated using the `stable-diffusion-2-1-base` model.

Automatic diffusion-based mitigation resulted in recovering 2 of 15 attack-success cases (Detection Recovery Rate of 13.33%) and preserving 13 of 15 control cases. For the five-image test, knowing the real location allowed the recovery of 2 of 4 targets for both the exact mask and the $+4$-pixel mask, compared to zero for the automatic localization path. At the same time, 2 attack-success cases remained unrecovered with known-location restoration, meaning that localization is not the only limiting factor.

Overall, the findings show that the transfer of adversarial patches should be considered with several detector-level metrics, and diffusion-based mitigation should be analyzed

separately for automatic localization and subsequent restoration/recovery steps.

VII. Discussion

Results from the experiment present a more comprehensive understanding of adversarial patches' behavior through the connection between the generation of adversarial patches, source-model vulnerability, transferability across architectures, victim-target effects, and patch diffusion mitigations. In this proposed experimental pipeline, there are six different detector models that have various tasks. In terms of attack generation, the model used as the source detector is Hyper-YOLO-N, while the victim models that are not seen before are Faster R-CNN ResNet-50 FPN V2, RT-DETRv2-S, D-FINE-S, YOLOv12-S, and Deformable DETR-R50. The adversarial patches utilized for the transfer attacks were created beforehand and remain the same.

A. Source-Model Behavior and Transferability

In total, Hyper-YOLO-N achieved an overall attack success rate (ASR) of 5.61% on the five original attack sets, with 69 successful attacks out of 1,229 eligible target instances. The attack set-wise ASRs are 5.33%, 7.39%, 5.00%, 4.53%, and 6.06% for A01–A05, respectively. This figure represents the behavior of the adversarial patches in the environment of the source model and therefore provides a baseline source measure rather than a transfer-target measure. Hyper-YOLO is a model with the MANet backbone and HyperC2Net neck, which features hypergraph-based feature computation and thus represents a different feature aggregation paradigm from that used in regular YOLO [10]. The source results show that the created adversarial patches successfully hindered the valid detection of target instances in the environment of the detector used to generate the attack sets.

As seen from the transfer results, only part of the source-model effect was preserved when the same adversarial images were tested on new, unseen detectors. This drop-off corresponds to the experimental setup because the five victim detectors were not included in the generation of the adversarial patches. As such, the transfer ASRs were 1.17% for Faster R-CNN, 2.38% for RT-DETRv2-S, 1.68% for D-FINE-S, 3.87% for YOLOv12-S, and 3.67% for Deformable DETR-R50. Therefore, the transferred effect was observed in all five victim models; however, the extent of the effect varied greatly depending on the detector architecture.

This difference between the source and victim results is important for analyzing the effectiveness of the adversarial patch in this case. While the patch can be highly effective in the source detector, it may lose some of its effectiveness after a change in the detector architecture. In turn, the observation of successful attacks in all five victim detectors shows that the adversarial effect is not specific to the source detector.

B. Faster R-CNN

The transfer Adversarial Success Rate (ASR) of Faster R-CNN was 1.17%, which is the lowest, with 16 adversarial samples out of 1,365 eligible targets. The algorithm works in a two-stage framework where the proposal of regions is made before the final classification and localization [12]. The weighted average of confidence drops was 0.012483, and the weighted average IoU drop was 0.001260, both numbers being quite low compared to other transfer victims.

This could be explained by the inability of the created adversarial pattern to change a correct Faster R-CNN detection to an incorrect one according to the described class-and-IoU matching criteria. It is possible that the connection between the adversarial patterns and the region proposal generation and detection occurs in a different way compared to the connection between the same adversarial patterns and the original detector. However, the current study cannot identify to which stage of the process this can be referred – either proposal generation, feature extraction, or classification. Thus, the transfer rate described above is an architecture-related effect and not a sign of robustness of the system.

The low drops in confidence and IoU indicate that for the cases where transfer occurred, the pattern caused only minor degradation according to the given criteria. Therefore, this detector could be used as a benchmark to show how a static patch loses its efficiency in transfer from a modern YOLO-based system to a two-stage detector.

C. RT-DETRv2-S

RT-DETRv2-S achieved an ASR of 2.38%, which is equivalent to 32 attacks out of 1,345 eligible target objects. RT-DETRv2 is a real-time DETR-based object detector designed to combine a transformer detection architecture with real-time performance, as discussed by Lv et al. [14]. Compared to Faster R-CNN, the transfer ASR is higher, suggesting that the same set of fixed adversarial patches resulted in more target-object invalidations for this detector.

It should be noted that one of the distinguishing features of RT-DETRv2-S is that its confidence and localization responses were more pronounced than the ASR suggested. The weighted mean confidence drop was 0.022492, while the weighted mean IoU drop was 0.006292, which is the highest aggregate IoU drop among all victim detectors. Therefore, the effect of the adversarial perturbations was not only expressed in the disappearance of target objects but also involved significant alteration of the existing detections in terms of either detector confidence or localization of the prediction, or both.

The difference between ASR and IoU behavior deserves special mention. While the relatively low ASR does not necessarily indicate a limited influence of the adversarial example on the detector, the response includes the degradation of existing predictions and not only their complete invalidation. In view of the fact that the experiment did not include an analysis of the transformer attention or feature activation, the architecture-based reason for this behavior remains unknown.

D. D-FINE-S

The ASR achieved by D-FINE-S is 1.68%, with 23 successful attacks out of 1,367 eligible targets. D-FINE represents

a highly granular DETR model with a focus on distribution refinement for bounding-box regression [15]. The weighted mean confidence drop stands at 0.019905, while the weighted mean IoU drop is equal to 0.004673.

In view of the comparatively low ASR together with the changes observed in confidence and IoU, one might assume that even though these transferred patches did not succeed in eliminating their targets, they still had some impact on the detector's performance. Such a conclusion can be made based on the results of RT-DETRv2-S, where degradation of the detector occurred despite the fact that the attack-success condition was not met.

The results of D-FINE-S also show that there is no reason to expect that detectors from a family using transformers have the same response to the same patch. Both RT-DETRv2-S and D-FINE-S showed a transfer effect, but their total ASR and localization responses were different. The difference can be explained by the different processing of features and bounding-box regression in the two models, but this assumption is not grounded because the current benchmark lacks component-level ablations.

E. YOLOv12-S

YOLOv12-S produced the highest attack success rate (ASR) of all five unseen victim detectors tested at 3.87%, with 49 successful attacks in total on 1,267 eligible targets. YOLOv12 is an attention-based variant of YOLO and represents a modern single-stage detector compared to the hypergraph-based source detector mentioned in Tian et al. [16]. YOLOv12-S demonstrated the largest weighted mean confidence decrease at 0.024935, while the weighted mean intersection-over-union (IoU) decrease was 0.002438.

A large ASR and confidence decrease are notable characteristics of YOLOv12-S. That means that the patches were relatively successful in altering the detection results and reducing the prediction confidence for this particular detector. At the same time, the IoU decrease was not the maximum, which means that the main effect of the patches on this detector is reduced prediction confidence and target invalidation rather than localization shifts among other valid detections.

Additionally, YOLOv12-S demonstrated the biggest class discrepancy in the transfer benchmark. The ASR for the *person* class is equal to 1.54%, while the ASR for the *car* class is 6.31%. As a result, there were more successful attacks causing target invalidation for car classes than for person classes within the used dataset. This should be taken as an observable characteristic of the current experiment, but not as a universal finding about adversarial patch vulnerability. The study did not independently control the size, appearance, posture, context, and position of objects and patches.

F. Deformable DETR-R50

Deformable DETR-R50 had a total ASR of 3.67%, where there were 48 successful attacks on 1,309 targets. Deformable DETR employs deformable attention in order to perform computations only at chosen spatial positions and gives a transformer-based alternative to traditional dense attention [13]. In the current experiment, the ASR of this victim was similar to that of YOLOv12-S, which means that the fixed adversarial patches could be transferred to this transformer-based detector successfully.

The weighted mean confidence decrease was 0.020449 and the weighted mean IoU decrease was 0.003289. The values provide evidence of significant changes in the detector behavior even though both were not the highest among the five victims. The significance of this result is the fact that it proves that considerable ASR could be reached in a detector architecture unusual for the source architecture, namely, in transformer-based detectors.

Therefore, the Deformable DETR-R50 result supports the main claim about the transferability of the adversarial patch effect: it is not restricted to a certain detector family. However, the dissimilarities between Deformable DETR-R50 and RT-DETRv2-S also prove that there is no uniform response of transformer-based detectors to the attack.

G. Comparison Across the Five Victim Detectors

Five different detector architectures show a wide variety of reactions to the identical transfer attack against their adversarial images. For instance, Faster R-CNN has the lowest ASR of 1.17%; D-FINE-S has 1.68%; RT-DETRv2-S has 2.38%; Deformable DETR-R50 has 3.67%; and YOLOv12-S has 3.87%. These disparities should be seen as variances in the transfer performance of detectors against the standardized benchmark attack, and not as an evaluation of their general performance and/or adversarial robustness ranking.

Furthermore, the comparison of the transfer performance of the five detectors shows that ASR alone cannot describe the extent of damage done to them. RT-DETRv2-S has a lower ASR than YOLOv12-S, yet it has a higher IoU change. At the same time, D-FINE-S has noticeable changes in confidence and localization despite its relatively low ASR. On the contrary, YOLOv12-S has the highest ASR among all detectors and the biggest weighted confidence loss but not the largest IoU change. Hence, this attack has different effects on detector behavior.

It might be possible that the above-described differences are caused by several design choices of the detector architectures, such as the backbone, feature fusion, attention mechanism, proposal generation, localization technique, prediction calibration, and so on. Nevertheless, in the current study, these features were not individually tested. The main conclusion that the experiment allows drawing is that there are distinct transfer reactions of detectors to identical adversarial images, which are associated with their different architectures.

H. Person and Car Target Behavior

In the set of victim detectors considered, there is a uniform trend in the class-level transfer performance. Car ASR exceeded person ASR in Faster R-CNN, D-FINE-S, YOLOv12-S, and Deformable DETR-R50, while RT-DETRv2-S achieved very similar results for the two categories. The combined

numbers are 0.45% vs. 1.87% for Faster R-CNN, 2.40% vs. 2.36% for RT-DETRv2-S, 1.34% vs. 2.01% for D-FINE-S, 1.54% vs. 6.31% for YOLOv12-S, and 2.43% vs. 4.92% for Deformable DETR-R50.

These observations indicate the potential impact of the target category on the transfer behavior. It is not the goal of the experiment to reveal the cause of the effect. Both classes include variability in object sizes, shapes, poses, appearances, and contexts, which is combined with the patch positioning constraint. Therefore, the results obtained at the class level can only serve as descriptive evidence obtained in relation to the COCO targets tested.

I. ASR, Confidence, and IoU as Complementary Measurements

Across the six different detector roles, the results showed that a single performance metric cannot be used to represent the impact of an adversarial patch. First, the ASR measures the number of previously valid targets that become invalid after the patching process, while confidence and IoU measure the degradation of those targets that are still valid.

This explanation is useful in the context of those detectors that show relatively low ASR but high variations in confidence and localization. For instance, RT-DETRv2-S had an ASR of 2.38% but showed the biggest change in IoU. On the same note, D-FINE-S showed only 1.68% ASR but had detectable changes in confidence and IoU. It means that the impact of an adversarial patch may be progressive, with the target's validity being degraded gradually without reaching the threshold of final validity loss.

Both-valid instances are necessary when it comes to confidence and IoU measurements because missing detections are not simply considered as having zero IoU or zero confidence. It means that the two metrics provide different information about how the patch affects the target detectors.

J. Implications of Diffusion-Generated Patch Content

Consequently, the importance of Stable Diffusion 2.1 Base in understanding the results of the attack is evident. In this case, the diffusion model is used to generate potential patch content, and Hyper-YOLO-N is used as a source detection environment. As a result, once the adversarial images have been generated, their content has been frozen for testing on other detectors. Thus, the results of the five victim models show the transferability of the generated adversarial patterns, and not the performance of the optimization process executed for each victim detector.

The fact that non-zero adversarial success rates (ASR) are achieved in all five victim models means that some disruptive potential of the generated patch content survived after the change of the source detector. However, taking into account the relatively low transfer ASRs compared to those of the source detector (Hyper-YOLO-N), it can be said that this disruption potential was not fully maintained after transfer. This statement is consistent with the overall development of adversarial patch generation and transfer research, which notes that these are two different problems [1]–[4].

Thus, it is not suggested that the adversarial patterns generated through diffusion are universally transferable. The point is that a transfer is performed in a single configuration of generation and placement of patterns. The differences between the five victim models indicate that the structure created through the diffusion model interacts with the detection process in individual detectors.

K. Automatic Diffusion-Based Mitigation

The mitigation experiments provide another point of view in testing how the affected area is detected and restored after the attack. In the Automatic DISPATCH experiment, a total of 30 images were tested, with 15 cases of successful attacks and 15 controls. Out of the 15 cases where the attacks were successful, 2 were recovered, providing a Detection Recovery Rate of 13.33%. Also, out of 15 controls, 13 were preserved, providing a Control Preservation Rate of 86.67%. The localization stage produced an Intersection-over-Union (IoU) of masks equal to 0.069325, mask precision of 0.090624, and recall of 0.775688. The average processing time was 114.24 seconds per image.

These figures suggest that it is possible to recover some of the attacked detections while preserving the majority of controls; however, automatic localization is still quite difficult. The fact that the recall is higher compared to precision indicates that the predicted suspicious regions include a part of the real patch region and some other areas around it. This discrepancy can affect restoration directly, since the diffusion model works on the basis of the predicted suspicious region.

The official automatic defense allows one to conclude about the existence of a two-stage dependency: to mitigate an attack, both good localization and restoration of this area are required. An error at the first stage can prevent the second stage from being performed properly, even though the restoration model is good enough.

L. Known-Location Restoration and Diffusion-Based Reconstruction

As another technique, known-location diagnosis enables us to evaluate the restoration module without automatic localization. In the case when the true coordinates of the patch are provided, the defense eliminates the risks arising from automatic localization and restores the damaged part of the image. Attack-success cases that could not be recovered via automatic processing were recoverable under the known-location setting, showing that the quality of automatic localization affects the results of the whole defense process.

The separate Stable Diffusion 2.1 Base Restoration test confirms this point qualitatively. From the image restored by the model, we can conclude that the diffusion mechanism is able to generate new information for the affected area after the patch removal. Nevertheless, it is unlikely that the generated region will be the same as the hidden one, since the original pixels that were covered by the patch cannot be used. Thus,

visual consistency and detection recovery should be considered as two related but separate goals.

It is especially relevant for the analysis of the diffusion-based defense mechanism. An image that looks plausible does not guarantee successful detection recovery, and vice versa. The defense needs to be evaluated in terms of detection recovery and restoration behavior.

M. Overall Discussion

The six-model evaluation demonstrates that the effectiveness of the proposed adversarial patches depends on interactions between the generation of the patches, source-detector representation and victim-detector architecture. Hyper-YOLO-N captured the source-model performance of the proposed attack, and the five unseen victim detectors showed how strong was the effect of that in case of the architectural modifications. According to the adopted protocol, Faster R-CNN demonstrated low transferability; RT-DETRv2-S and D-FINE-S demonstrated medium transferability with some decrease in surviving detections; and YOLOv12-S and Deformable DETR-R50 demonstrated relatively high target invalidation rate. At the same time, the changes in the confidence and IoU metrics were inconsistent with the observed pattern of ASR, indicating the multidimensional character of adversarial effects.

Furthermore, one could conclude that the transferability of diffusion-based adversarial patches cannot be considered architecture-independent. The same visual perturbations led to different ASR and degradation patterns of five victim detectors, and source Hyper-YOLO-N result stood out because of the direct connection with the attack generation environment. It can be supposed that the analysis of class-specific data shows that target characteristics play a certain role in the obtained results. However, the adopted benchmark does not allow to determine the exact contribution of target category in the obtained effect.

Lastly, the performed mitigations demonstrate that the diffusion-based recovery contributes to the patch removal. But full defense strongly relies on the localization process. The automatic recovery provided by the official benchmark achieved partial success while preserving all controls, and the known location diagnostics allowed to achieve some extra recoveries. All together, the presented results imply that the adversarial patches research should be considered from the integrated point of view including patch generation, attack effectiveness of the source model, cross-detector transfer, degradation at the target level, localization, diffusion-based recovery and detector re-evaluation.

VIII. Limitations

Despite the comprehensive experimental study on adversarial patch transferability through diffusion as well as mitigation via diffusion in this research, there are certain limitations that limit the generalization of the results of this experiment. The limitations mainly revolve around the target classes and image domain used in this experiment, the patch generation and transfer methodology, the size of the mitigation experiment, and the computing environment.

A. Dataset and Target-Class Scope

The study used a subset of images from the MS COCO database and only considered attacks and transfers for two specific targets, i.e., the categories “person” and “car.” Thus, it would be erroneous to generalize the observed attack and transfer behavior to all other object categories that differ in their visual appearances and sizes. As a result, the specific observations made here are limited to the specific COCO targets.

Also, the testing was done using digital images alone. The attacks and patched images were analyzed assuming no physical limitations such as printing errors, changing illumination, varying viewpoints, distortions, movements, and so on. This means that the findings are limited to digital attacks.

B. Patch Generation and Attack Optimization

The evaluation was carried out on a selected sample of images from the MS COCO dataset, which contained only two target object classes: person and car. Hence, the obtained attack and transfer effects cannot be taken as an example of the behavior of the adversarial attacks for any other categories. The adversarial patches used in this research were candidate diffusion-based adversarial patches that have been developed within the Hyper-YOLO-N adversarial attack framework. Adversarial patches have been used as a fixed input to evaluate their transferability to other victim detectors. Neither patch generation nor optimization for the specific victim detector has been applied to Faster R-CNN, RT-DETRv2-S, D-FINE-S, YOLOv12-S, or Deformable DETR-R50.

Hence, the reported Transfer ASR value should not be regarded as an indicator of the maximal possible ASR that can be achieved if we use another attack configuration for generating a new patch for each victim detector. Thus, these results should not be considered as a complete characterization of the adversarial robustness of the evaluated detectors.

Besides, the created patches belong to one Adversarial configuration: patches are of a fixed size and placed at fixed positions on the attacked image. Different patches sizes, different shapes, various optimizations, and generation procedures could result in different behavior of the patches. Thus, this study characterizes only one specific adversarial attack scenario instead of all possible adversarial attacks.

The observation concerning the attack on specific categories is valid only for those objects that have been evaluated on COCO images and can’t be extrapolated to other categories with other visual properties. Finally, this research is done in the digital domain. Patches were generated and tested without any physical world constraints like printing defects, changes in illumination, viewpoint, motion, and deformations.

C. Attack-Set Sampling and Cross-Model Interpretation

The attack set samples form different subsets of sampled images. Thus, variations in measured attack-level ASR may

come from both different adversarial perturbations and different target instances, shapes of objects, scenes of images and other image properties. Hence, discrepancies between A01–A05 should not be interpreted as an absolute ranking of patch effectiveness.

The similar issue applies to inter-model comparison as well. Even though the transfer attack utilizes exactly the same set of adversarial images for all the victim detectors, the number of clean detections eligible for the test varies across models. In turn, ASR is calculated based on clean detections according to the common class-and-IoU matching rules, not on the number of images. Such methodology provides a unified test of target properties; however, discrepancies in eligible detection pools may affect overall statistics as well.

Finally, the current research does not include feature-level or component-wise analysis of six detectors under test. Therefore, even though architectures show dissimilar ASR, confidence and IoU results, no particular element of an architecture (backbone, attention model, proposal step, feature fusion component, or regression model) can be claimed as the reason for these discrepancies.

D. Limited Mitigation Evaluation

The experiments for mitigation were performed on much smaller samples than the 1,500 images used for the transfer experiment. Officially, the Automatic DISPATCH test analyzed 30 images, 15 images in the group of attacks that have succeeded and 15 images in the control group. Hence, the obtained results of the Detection Recovery Rate (DRR) and Control Preservation Rate (CPR) must be viewed as statistics about the subset tested for mitigation rather than about the overall defense performance with respect to the entire attack dataset.

The mitigation subset is also too small to ensure sufficient stability of class-specific recovery rates. More specifically, there is a low number of samples with attacks succeeding on people and cars. Thus, a more significant mitigation benchmark must be used in order to test the stability of the recovery pattern.

E. Automatic Localization Limitation

The official Automatic DISPATCH experiment assumes an estimate of the suspicious patch region before diffusion restoration. In the automated framework, the checkerboard reconstruction, L2 difference map, smoothing, and KMeans clustering operations are used for obtaining the predicted suspicious mask. The defense is not provided with the ground-truth patch coordinates, and no clean pixels are compositely incorporated into the restoration results.

Measuring localization quality demonstrates one significant drawback at this step. The official experiment produces the average IoU of 0.069325 for the adversarial mask, where mask precision equals 0.090624, and mask recall equals 0.775688. The difference between recall and precision suggests that predicted masks are likely to cover a large part of the perturbed region but also include some unrelated surroundings. This localization error may influence the restoration step, as the diffusion is performed based on the estimated suspicious region.

Thus, the final recovery rate shows the combined performance of localization and restoration, not just the restoration model itself. Therefore, improvements of the generative restoration module may prove ineffective if the localization step performs poorly.

F. Known-Location Diagnostic Assumption

This diagnostic experiment for restoration was performed using the known location of the patches, i.e., the correct patch locations were given to the defender. Such an arrangement makes the distinction between localization errors and restoration errors possible, although it cannot be viewed as an instance of a truly blind defense situation.

In real-world applications, the defender will not have any knowledge of the correct patch locations in advance. Therefore, the recovery achieved in this case cannot be considered as a measure of the performance of an automatic patch detector. It serves only as an example of recovery in a situation where localization errors are small.

G. Diffusion Restoration Fidelity

In terms of the restoration phase based on the diffusion model, the intention is to restore the region damaged by the adversarial patch, but the original pixels covered by the patch cannot be extracted from the damaged image. As a result, the obtained restoration will be a synthetic approximation rather than a real restoration of the original image.

This provides an important difference between visual restoration and detector restoration. While a restored image may look like a plausible one and at the same time provide a different prediction for the detector compared to the original image, a detector can recognize the target in the image while the generated local region does not perfectly match the original region. Thus, it is impossible to obtain a reliable reconstruction of the original image in the current experiment.

The qualitative experiment on restoration with the Stable Diffusion 2.1 Base model should be understood through this limitation as well. The experiment shows how a diffusion model can generate a replacement local region, but it does not prove that this generated region looks like the hidden region.

H. Computational and Implementation Constraints

However, mitigation through diffusion requires higher computational resources than the object-detection inference process. In the Automatic DISPATCH experiment, the average time for an image was 114.24 seconds. These additional computational resources hinder the direct application of the current mitigation process for large-scale or real-time object detection.

Moreover, there were restrictions related to the computing conditions under which the Deformable DETR-R50 model was evaluated. The pre-trained weights of this model were evaluated in Windows conditions, whereas the `ms_deform_attn`

CUDA operations provided by the official library were not compiled on the host machine. These conditions of the implementation must be considered when interpreting the detector performance because of the unoptimized experimental execution conditions.

I. Scope of the Defense Evaluation

The mitigation strategy in this case study is limited to diffusion-based restoration as well as the localization-restoration pipeline outlined through the selected adversarial attack defenses. It is not a complete comparison against all existing adversarial patch defenses. Therefore, the mitigation results are to be interpreted as a scientific experiment involving the selected diffusion-based mitigation strategy and not all existing defense techniques.

In conclusion, the above-mentioned limitations outline the range of the current study. This study offers a scientific benchmark test for the transferability of diffusion-generated adversarial patches for six different detectors. However, conclusions related to attacks in the physical world, victim-dependent optimization of adversarial patches, performance at scale, and detector-independent robustness require more scientific experiments.

IX. Conclusion

The current work aims at empirically investigating the transferability of diffusion-generated adversarial patches and the potential of diffusion-based restoration for the mitigation of such attacks on target detection. As the source detector, Hyper-YOLO-N is utilized, and five unseen victim detectors—Faster R-CNN, RT-DETRv2-S, D-FINE-S, YOLOv12-S, and Deformable DETR-R50—are examined based on the same fixed adversarial image dataset. For the source detector, the attack success rate (ASR) was measured to be 5.61%, while for the transfer victims, the ASRs ranged from 1.17% to 3.87%.

These results demonstrate that diffusion-generated adversarial patches are transferable to different detector architectures and cause architecture-dependent attacks. However, further results show that the impact of the adversarial patches cannot be described only through ASR, since the confidence and IoU also vary depending on the attack. Moreover, class-specific results of the analysis showed that the response of the detector varies depending on the targeted object class. Person and car targets were selected for testing, and differences in response were observed between victim detectors.

Hence, the results obtained demonstrate the necessity of a multi-measure approach to analyzing cross-detector adversarial attacks and mitigation. To examine the possibility of restoring the detection after the adversarial patch application, the Automatic DISPATCH experiment was conducted and yielded a Detection Recovery Rate of 13.33% while maintaining a Control Preservation Rate of 86.67% for the 15 attack-success cases and 15 control cases. While the automatic localization performed poorly, diagnostics of mask overlap with the known adversarial region showed that successful localization plays a key role in achieving restoration success. The Stable Diffusion 2.1 Base experiment shows the possibility of visually successful restoration but not necessarily the restoration of the exact pixels hidden by the adversarial patch.

In conclusion, the current work shows the capability of diffusion-based adversarial patches to cause transferable adversarial attacks and demonstrates the potential of diffusion-based restoration techniques in mitigating such attacks after adversarial patch localization. The presented results show the need for a comprehensive framework of evaluation that includes the generation of the patches, attack by the source detector, transfer, target-level degradation, localization, and restoration followed by the new detection. In future work, the framework can be expanded to other objects and conditions, victim-targeted optimization, a larger restoration dataset, and more accurate localization.

Acknowledgment

We would like to express our sincere gratitude to our supervisor, Azizur Rahman Anik, Lecturer, Department of Computer Science and Engineering, for his valuable guidance, continuous support, and constructive feedback throughout this research work. We are also grateful to the Department of Computer Science and Engineering, United International University, for providing the academic environment and necessary support for conducting this study. Finally, we would like to thank our family and friends for their encouragement and support throughout the completion of this research.

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